Porosity effects on dielectric properties

"How does porosity affect dielectric properties in SiO2-MoS2 TENG materials?" *Generated by LeapSpace*

Porosity in **SiO_2–MoS_2 TENG materials** mainly shifts the dielectric response through a balance of **more interfacial polarization and trap sites** versus **more air voids and easier leakage/low permittivity**; the net effect is usually best when porosity is **controlled, closed, and moderate rather than maximized** [48] [91] [92] [93] [94] [95].

How porosity changes dielectric properties

Porosity feature	Dielectric effect	TENG relevance
More pore surface / more interfaces	Increases **interfacial polarization** and trap density, which raises effective polarization and charge storage [77] [91] [96] [97]	More trapped charge and stronger surface charge density
Air-filled voids	Lowers the **effective dielectric constant** because air has very low permittivity; porous PS/MWCNT films showed permittivity drops as pore size increased [75]	Can hurt charge-storage capacity if porosity is excessive
Small, closed pores	Can reduce dielectric constant and loss while preserving barrier behavior; closed porous structures are linked to lower dielectric loss and better water resistance [95]	Helpful for stable operation in humid environments
Large or open pores	Can increase pore connectivity and reduce dielectric performance by increasing losses or leakage pathways; large pores reduce free surface area and can reduce dielectric loss relative to smaller pores in porous dielectrics [92] [94]	Riskier for long-term charge retention
Hierarchical porosity	Combines high surface area with charge-transport pathways, strengthening dielectric response when leakage is controlled [98]	Can improve output if the shell still remains insulating enough

What this implies for SiO_2–MoS_2 specifically

- **SiO_2-rich porosity tends to help dielectric trapping** because SiO_2 domains can act as localized charge-trapping sites and increase the dielectric constant through interfacial polarization [91] [99].
- **MoS_2 contributes interfacial polarization and charge trapping**, so porosity is most beneficial when it **increases contact area without breaking shell continuity** [100] [101].
- In a core-shell design, the best-performing structure is likely **not the most porous one**, but the one with **controlled, well-distributed pores** that preserve the insulating/charge-retaining role of the SiO_2–MoS_2 interface [75] [93] [95].
- The dielectric loss trend is also important: porosity can **lower permittivity** by adding air, yet **increase charge trapping** by adding interfaces; this is why a **moderate porosity window** is the most defensible target for TENGs [48] [75] [102].

Bottom line

For **SiO_2–MoS_2 TENG materials**, porosity is useful when it is engineered to create **more interfacial polarization and trap sites** while keeping the structure **closed, continuous, and leakage-resistant**; if porosity becomes too open or too large, the dielectric constant drops and charge retention can worsen [48] [75] [92] [95].

Confidence level: **Medium** — the retrieved evidence strongly supports how porosity affects dielectric properties in porous dielectrics and TENG composites, but it does not include a direct pore-by-pore dielectric sweep for one exact SiO_2–MoS_2 TENG formulation.

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